

# Three Generations of Healthcare IT: From the Digital Record to the Computable Care Process

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## Abstract

Objective. Healthcare information technology is usually organized by the technologies it adopts. We instead organize it by the unit of information a system makes computable, and describe an emerging computational layer whose object is patient-specific clinical intent.

Approach. We give criteria for a computational layer, derive three (record, clinical state, and a proposed layer of clinical intent), and formalize the Actionable Clinical Record (ACR) as the atomic object of the third layer.

Discussion. The framework distinguishes prescribed, observed, and intended process, and identifies recovery of intended process from communication as the capability existing standards do not provide. The ACR is complementary to FHIR workflow resources, computer-interpretable guidelines, and process mining; a companion feasibility study supports technical tractability.

Conclusion. Computable clinical intent is a coherent research direction; the ACR, its maturity ladder, and an executable-correctness evaluation framework are reusable constructs that subsequent work can extend, evaluate, or falsify.

**Keywords:** electronic health records; clinical workflow; actionable clinical record; clinical intent; natural language processing; large language models; FHIR; care-process automation

## 1. Introduction

Healthcare information technology can be analyzed more durably through the computational objects it supports than through the technologies it adopts, which change rapidly.<sup>[1,2]</sup> Organized this way, the field’s history is compact: the first generation made the clinical *record* computable, the second the clinical *fact*, and a third, which we propose, makes patient-specific clinical *intent* computable from communication. We use *generation* for the historical emergence of a new computational layer; the layers accumulate rather than replace one another.

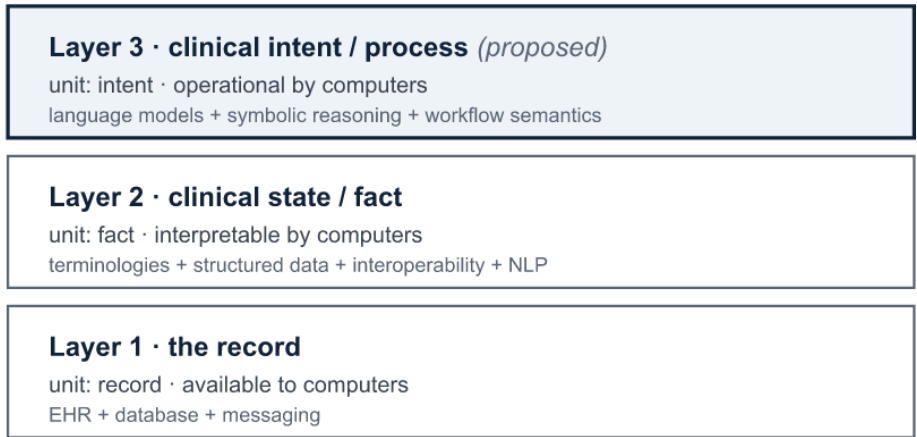
A limitation persists after the first two layers are in place. When a clinician writes “repeat the complete blood count in two weeks” or “refer to cardiology if symptoms persist,” the instruction is decisive yet captured only as narrative text; its action, timing, condition, and responsible actor remain outside the computable substrate. The cost is measurable: test-result follow-up in ambulatory care often fails,<sup>[3]</sup> only about one-third of specialist referrals reach a documented completed visit,<sup>[4]</sup> and such loop-closure failures contribute to diagnostic error and avoidable harm.<sup>[5,6]</sup>

This Perspective offers a framework and a vocabulary rather than a new empirical study: criteria for a computational layer and a three-layer organization of the field (Section 2); a formal definition of the Actionable Clinical Record (ACR) as the atomic object of the third layer (Section 5); an executable-correctness evaluation framework with a companion feasibility study (Section 7); and a research agenda (Section 8).

## 2. A Framework of Computational Layers

We distinguish a computational layer from an incremental trend by five markers: a new *atomic unit of computability*; an *external driver*; an *enabling technology*; a new *class of computation* the unit permits; and a *residual limitation* that motivates the next layer's object. The layers are *cumulative*, each subsuming those beneath it (Figure 1), and differ in epistemic status: the first two are *retrospective abstractions* over a well-documented history of electronic health records,<sup>[1,2]</sup> whereas the third is a *prospective hypothesis*. The lens is complementary to the data–information–knowledge hierarchy<sup>[7]</sup> and to digital-maturity models such as EMRAM,<sup>[8,9]</sup> which measure how much capability an institution has adopted; our concern is *which* clinical object becomes computable (Table 1).

each layer subsumes the one below



**Figure 1.** The three computational layers of healthcare IT, organized by the unit of information each makes computable. Layers accumulate rather than replace; layer 3 is proposed. Alt text: three stacked boxes, record at the base, clinical state above it, and clinical intent at the top, with an upward arrow indicating that each layer subsumes the one below.

| Axis                  | Layer 1: Record                    | Layer 2: Clinical state                                        | Layer 3: Clinical intent (proposed)                     |
|-----------------------|------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|
| Unit of computability | Clinical record                    | Discrete clinical fact                                         | Intended clinical action (ACR)                          |
| Question answered     | What was documented?               | What is true about the patient?                                | What is supposed to happen?                             |
| External driver       | Adoption policy and reimbursement  | Interoperability and quality measurement                       | Care continuity and patient safety                      |
| Enabling technology   | Transactional databases; messaging | Terminologies; structured data; interoperability; clinical NLP | Language models; symbolic reasoning; workflow semantics |
| Class of computation  | Store, retrieve, exchange          | Query, reason, predict                                         | Schedule, coordinate, monitor, close                    |
| Canonical output      | Document                           | Coded fact                                                     | Actionable Clinical Record                              |
| Residual limitation   | Meaning remains in narrative       | Intent remains in communication                                | Safe, reliable execution                                |

**Table 1.** The three computational layers against the framework markers. Layers 1 and 2 are retrospective; layer 3 is proposed. The residual limitation of each layer motivates the unit of the next.

### 3. The First Two Layers: Record and Clinical State

The first layer digitized the clinical record and its transactions, replacing the paper chart with electronic capture, storage, retrieval, transmission, and operations such as order entry and results routing. In the United States, HITECH and Meaningful Use moved adoption to near-universal within a decade,<sup>[10,11]</sup> extending the older problem-oriented-

record ideal of an organized, auditable chart.<sup>[12]</sup> Its enabling technology, the transactional database, stored and moved information without making its clinical meaning computable across systems.

The second layer turned the record into computable clinical *state*. Its enabling stack was broad and cannot be reduced to a single technique: structured data entry, controlled terminologies,<sup>[13]</sup> computerized order entry and coded lists, clinical data warehouses, and interoperability standards<sup>[14,15]</sup> were as consequential as language processing, which was one route by which narrative facts entered the semantic layer (MedLEE and cTAKES parsing reports into coded concepts).<sup>[16,17]</sup> These enabled the first broad wave of computation *on* clinical data: decision support, quality measurement, registries, and research.<sup>[18]</sup> A tension remained: encoding meaning into structured fields captures facts, while much of the reasoning, and nearly all of the intended action, stays in narrative.<sup>[19]</sup>

#### 4. The Remaining Computational Gap

The first two layers made clinical *state* computable but not the *intended actions* that constitute care. Healthcare IT can represent process, but only once a person or system has structured it: FHIR represents patient-specific intent through *CarePlan*, *ServiceRequest*, and *PlanDefinition*;<sup>[14]</sup> computer-interpretable guidelines encode executable protocols;<sup>[20,21]</sup> and process mining reconstructs pathways from event logs.<sup>[22,23]</sup> These paradigms are complementary to the proposed layer and often provide the infrastructure to operationalize its outputs. The gap is isolated by distinguishing three kinds of process information (Table 2): *prescribed* (what should generally happen), the domain of guidelines and decision support; *observed* (what actually happened), the domain of

98 records and process mining; and *intended* (what is supposed to happen for a specific pa-  
 99 tient), expressed primarily in communication. Existing paradigms address the first two.  
 100 The missing capability, and the subject of the third layer, is recovering intended process  
 101 from ordinary communication and converting it into computable, executable form.

| Process type | Meaning                                           | Typical source         | Existing paradigm                                   |
|--------------|---------------------------------------------------|------------------------|-----------------------------------------------------|
| Prescribed   | What should generally happen                      | guideline / protocol   | computer-interpretable guidelines; decision support |
| Observed     | What actually happened                            | event log / EHR        | process mining                                      |
| Intended     | What is supposed to happen for a specific patient | clinical communication | proposed third layer                                |

102 **Table 2.** Three kinds of process information. The proposed third layer targets intended process,  
 103 which the prescribed and observed paradigms do not recover from natural communication.

## 104 5. Computable Clinical Intent

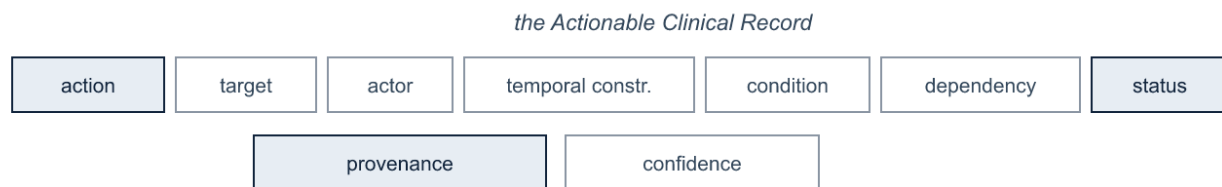
105 We propose that the third layer makes patient-specific clinical intent computable by re-  
 106 covering it from communication as a structured, executable representation. We use one  
 107 terminology throughout: clinical *communication* expresses clinical *intent*; an intent de-  
 108 composes into intended *actions*; each action is an *Actionable Clinical Record*; a set of  
 109 ACRs and related events is a clinical *workflow*; and the enacted sequence over time is  
 110 the *care process*.

### 111 5.1 The Actionable Clinical Record

112 Definition 1 (Actionable Clinical Record). An Actionable Clinical Record is a source-  
 113 grounded representation of a patient-specific intended clinical action, with the semantic  
 114 attributes required to interpret, execute, monitor, or audit it: the tuple  
 115  $ACR = \langle \text{action, target, actor, temporal constraint, condition, dependency, status, prove-}$   
 116  $\text{nance, confidence} \rangle$

in which *action*, *status*, and *provenance* are required, and the remaining attributes are populated when the communication supports them.

The attributes are defined semantically (Figure 2). Beyond *action*, *target* (linked where possible to a layer-2 coded concept), and the responsible *actor*, the *temporal constraint* is a normalized time semantics rather than a date alone, admitting absolute (“within six months”), relative (“before the next visit”), event-anchored (“after finishing antibiotics”), and recurring forms; *condition* is a clinical trigger; *dependency* encodes ordering; *status* is the rung on the maturity ladder; *provenance* grounds the record in its source span, speaker, and encounter; and *confidence* is a calibrated per-attribute uncertainty routing uncertain records to review.

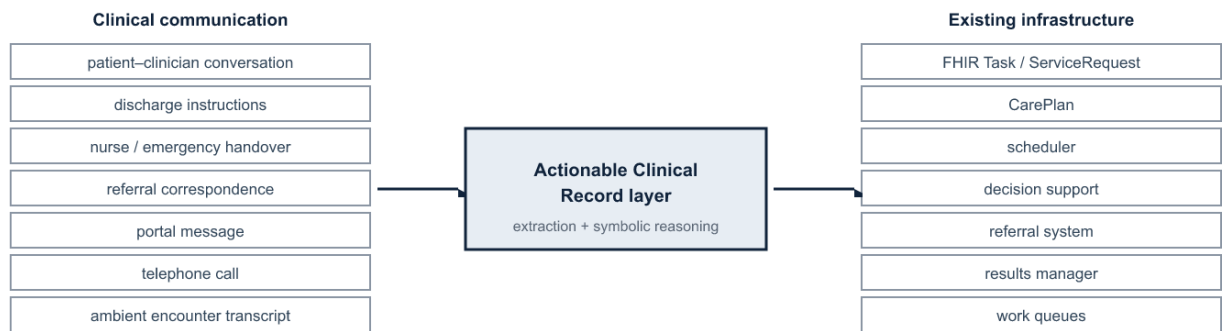


**Figure 2.** The ACR schema. Alt text: nine labeled boxes forming the ACR tuple; *action*, *status*, and *provenance* are shaded to mark them as required; *target*, *actor*, *temporal constraint*, *condition*, *dependency*, and *confidence* are outlined as optional.

The third layer is a capability ladder, and naming its rungs prevents over- and under-claiming. Recognizing that “complete blood count” appears in a note belongs to layer 2; extracting “repeat complete blood count” is a partial step; linking it to “two weeks” and a responsible actor produces an ACR; mapping it to a computed due date and a workflow object makes it executable; and detecting completion or escalation closes the loop. We label the rungs *mentioned*, *interpreted*, *actionable*, *executable*, *monitored*, and *closed*; the ladder doubles as the ACR’s *status* attribute and distinguishes an *actionable* record from an *executable* one carrying enough resolved information to act on automatically.

## 5.2 From communication to workflow

The third layer is a bridge rather than a replacement electronic health record (Figure 3). Its inputs are the range of clinical communication (discharge instructions, patient–clinician conversation, handovers, referral correspondence, portal messages, telephone calls, ambient transcripts); its output is a set of ACRs that maps onto infrastructure the field already has, including FHIR *Task*, *ServiceRequest*, and *CarePlan*, schedulers, referral systems, and results management.<sup>[14]</sup> Existing systems execute instructions once structured; the capability the third layer adds is recovering them when they originate as natural communication.



**Figure 3.** The ACR as the layer between clinical communication and existing workflow infrastructure. Alt text: communication sources on the left feed a central Actionable Clinical Record layer, which maps to existing infrastructure such as FHIR resources, schedulers, and referral and results systems on the right.

Follow-up is the most-documented case, but it is one of a broader class of communication-derived process objects sharing the ACR representation: conditional escalation (“return immediately if fever develops”), medication transitions (“reduce the dose after seven days”), handoff commitments, referrals, pending-result responsibility, and patient self-management. Recent work on patient messages, incidental-finding follow-up, and ambient documentation indicates these channels routinely carry clinically consequential actionable content.<sup>[24]</sup>

### 5.3 Relation to existing paradigms

Two mature lines can be mistaken for the third layer. Clinical NLP produces *facts*, recording that a concept appears;<sup>[16,17,25]</sup> the third layer produces an executable *action*, at a reliability sufficient to act on rather than only index. Advances in language models improve the feasibility of recovering the relevant relations, but improving NLP is not, by itself, a layer transition. Computer-interpretable guidelines encode generic, population-level protocols;<sup>[20,21]</sup> the third layer recovers patient-specific records from a particular encounter. The two are complementary: guidelines state what should generally happen; the third layer captures what a clinician instructed for this patient.

### 6. Reliability and Architecture

A more capable text generator does not remove the requirements the third layer imposes. Auditability, verifiable temporal correctness, and a calibrated signal of when to defer are properties of the output contract, not gaps in model capability: a system acting on patient trajectories must establish that a computed date is correct and trace it to its source, however fluent the model. General-purpose language models interpret clinical language well but are less reliable where the layer is most demanding, in binding an action to its time and normalizing it into an executable schedule without omission or fabrication.<sup>[26,27]</sup>

These requirements motivate a hybrid decomposition in which learned components interpret and explicit components enforce verifiable structure.<sup>[28]</sup> Because some workflow semantics (calendar arithmetic, dependency ordering, status transitions) are formally specified, learned extraction can supply candidate actions and arguments<sup>[29]</sup> while deterministic reasoning resolves times and computes due dates<sup>[30,31,32]</sup> and selec-

tive prediction routes uncertain cases to a clinician.<sup>[33]</sup> The reliability requirements can be stated as invariants: every acted-upon record traces to a source span, speaker, and encounter; deterministic semantics are computed, not generated; each inferred attribute carries a calibrated confidence; the system may abstain rather than guess; higher-consequence actions receive tighter human review;<sup>[33]</sup> an immutable trace links communication to action for audit;<sup>[34]</sup> and extracted content stays distinct from inferred content. Reliability, not scale, is the governing constraint.

## 7. Evaluation Framework and Feasibility

Because the third layer acts, it should be evaluated on *executable correctness* rather than text overlap. We propose an evaluation framework for ACR recovery whose measures are reported separately: action detection; argument extraction; action–time, condition, and actor linking; temporal-normalization error; whole-record exact match; unsupported- and omitted-action rates; calibration and selective-risk behavior; source-provenance accuracy; and loop-closure outcome. This framework, rather than any single score, is the contribution we intend others to adopt.

A companion study offers a formative feasibility test of one minimal task, follow-up-instruction extraction, evaluating a hybrid neural-symbolic pipeline on a controlled benchmark against generative baselines.<sup>[35]</sup> On the action–time pairing the executable representation requires, the hybrid pipeline substantially outperforms general-purpose language models, which recognize actions accurately but bind them to times unreliably. This establishes technical feasibility for one task; it does not, by itself, evidence the framework, whose value rests on the constructs it defines and their evaluation across tasks.

## 8. Research Agenda

The framework defines a program in four domains. **Representation:** a consensus ACR ontology, with an extensible taxonomy of actions, attribute semantics, constraint types, provenance, and reconciliation when communications revise or revoke a plan. **Inference and reliability:** ACR recovery from multilingual and multimodal communication, temporal reasoning, cross-message reconciliation, calibrated abstention, and distinguishing patient from clinician intent. **Evaluation:** authentic-data benchmarks beyond synthetic corpora, the executable-correctness framework of Section 7 adopted across tasks, and cross-institution and cross-language generalization. **Integration and impact:** ACR-to-FHIR mapping and write-back into scheduling, results-management, and referral systems,<sup>[14,20]</sup> human-factors design of the review queue, regulatory classification, and whether closed-loop operation improves clinical outcomes.

Moving from synthetic corpora to de-identified real-world notes across languages and documentation styles is the central empirical risk, and safe deployment presupposes the invariants of Section 6. The proposal is falsifiable: machine-actionable use of FHIR workflow resources should grow relative to free-text follow-up; evaluation should shift toward executable-correctness measures; reliable systems should expose source-linked, auditable records with selective review rather than end-to-end generation; and ambient documentation should extend from notes to tasks and orders.<sup>[24,36]</sup> Failure of these trends would count against it.

## 9. Conclusion

Healthcare IT has made the record and then the clinical fact computable; we propose patient-specific clinical intent as the next layer, the layers accumulating rather than re-

placing one another. What characterizes it is not generating clinical text but recovering, from ordinary communication, source-grounded and verifiable representations of intended action, formalized as the Actionable Clinical Record. The layer supplies the intent that lets natural communication reach existing workflow infrastructure, under reliability formulated as an explicit system property. We offer the framework, the ACR, its maturity ladder, and the evaluation framework as reusable constructs that subsequent work can operationalize, evaluate, extend, or falsify.

### Data Availability

No new data were generated for this article. The feasibility results discussed in Section 7 derive from a companion study;<sup>[35]</sup> details of that benchmark are reported therein.

### Author Contributions (CRediT)

A. Apartsin: Conceptualization, Methodology, Writing – original draft. Y. Aperstein: Conceptualization, Supervision, Writing – review and editing.

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### Conflicts of Interest

None declared.

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